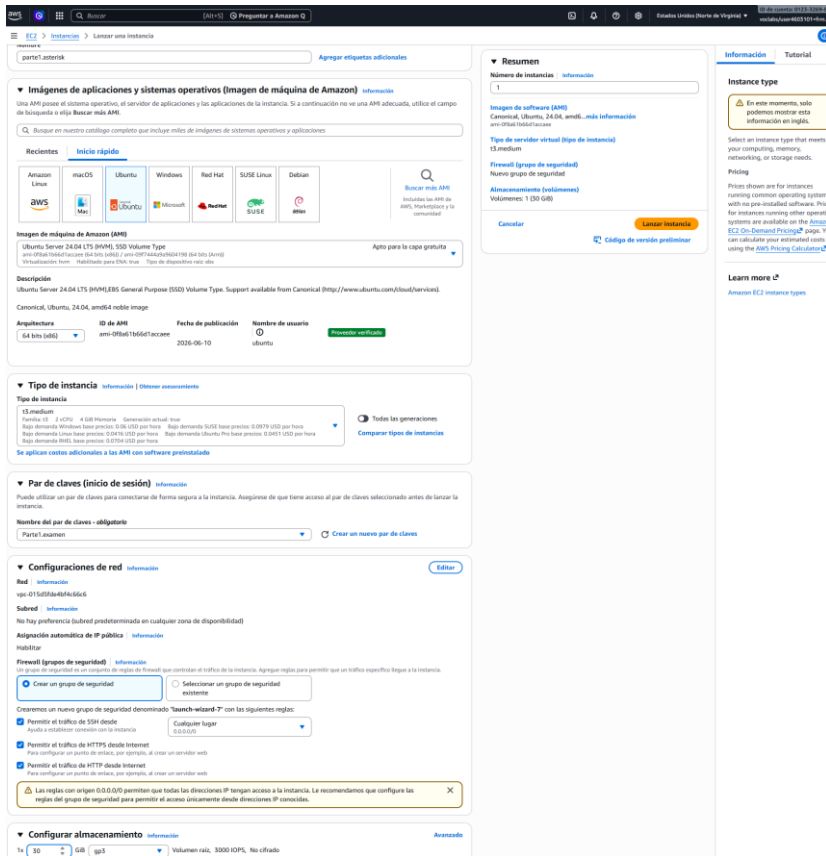




PARTE 1 EXAMEN

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Resumen

- Imagen de software (AMI)**: Canonical, Ubuntu, 24.04, amd64, [más información](#)
ami-0fba61b6d1ecace
- Tipo de servidor virtual (tipo de instancia)**: t3.medium
- Firewall (grupo de seguridad)**: Nuevo grupo de seguridad
- Almacenamiento (volúmenes)**: Volúmenes: 1 (20 GiB)

Configuraciones de red

Subred: vpc-015f1b6d4b6d1ecace

Asignación automática de IP pública: No hay preferencia (subred predeterminada en cualquier zona de disponibilidad)

Firewall (grupos de seguridad): ☒ Crear un grupo de seguridad ☐ Seleccionar un grupo de seguridad existente

Crear un nuevo grupo de seguridad denominado "trench-wizard-7" con las siguientes reglas:

- ☒ Permitir el tráfico de SSH desde cualquier lugar (0.0.0.0/0)
- ☒ Permitir el tráfico de HTTP desde Internet
- ☒ Permitir el tráfico de HTTP desde Internet

Configurar almacenamiento

1x **30** **GiB** **gp3** **Volúmenes raíz, 3000 IOPS, No cifrado**

```
ubuntu@ip-172-31-18-142:~$ wget https://raw.githubusercontent.com/nicsanchezr/NEW-CUY5132-DUOC/refs/heads/main/Experiencia-1/install_pbx.sh
--2026-07-14 16:44:48-- https://raw.githubusercontent.com/nicsanchezr/NEW-CUY5132-DUOC/refs/heads/main/Experiencia-1/install_pbx.sh
resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Content-Length: 30865 (30K) [text/plain]
Saving to: 'install_pbx.sh'

install_pbx.sh      100%[=====>]  30.14K  --.-KB/s    in 0s

2026-07-14 16:44:48 (103 MB/s) - 'install_pbx.sh' saved [30865/30865]

ubuntu@ip-172-31-18-142:~$
```

2. Preparar la estructura

En la máquina Asterisk ejecuta:

```
sudo apt update
```

```
sudo apt install -y python3 sqlite3 sox curl
```

```
ubuntu@ip-172-31-18-142:~$ sudo apt install -y python3 sqlite3 sox curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.12.3-0ubuntu2.1).
python3 set to manually installed.
```

```
sudo install -d \
```

```
-o asterisk \
```

```
-g asterisk \
```

```
/var/lib/asterisk/citas
```

```
ubuntu@ip-172-31-18-142:~$ sudo install -d \
-o asterisk \
-g asterisk \
/var/lib/asterisk/sounds/custom/citas
ubuntu@ip-172-31-18-142:~$
```

```
sudo install -d \
```

```
-o asterisk \
```

```
-g asterisk \
```

```
/var/lib/asterisk/sounds/custom/citas
```

```
sudo touch /var/log/asterisk/citas-agi.log
```

```
sudo chown asterisk:asterisk \
```

```
/var/log/asterisk/citas-agi.log
```

```

/var/lib/asterisk/sounds/custom/citas
ubuntu@ip-172-31-18-142:~$ sudo touch /var/log/asterisk/citas-agi.log
ubuntu@ip-172-31-18-142:~$ sudo chown asterisk:asterisk \
/var/log/asterisk/citas-agi.log
ubuntu@ip-172-31-18-142:~$

collecting protobuf>=4.25.8 (from onnxruntime<2,>=1->piper-tts)
  Downloading protobuf-7.35.1-cp310-abi3-manylinux2014_x86_64.whl.metadata (595 by
es)
  Downloading piper_tts-1.4.2-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.m
anylinux_2_28_x86_64.whl (13.8 MB)
----- 13.8/13.8 MB 129.2 MB/s 0:00:00
  Downloading onnxruntime-1.27.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x8
6_64.whl (18.7 MB)
----- 18.7/18.7 MB 100.3 MB/s 0:00:00
  Downloading pathvalidate-3.3.1-py3-none-any.whl (24 kB)
  Downloading numpy-2.5.1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.wh
l (16.7 MB)
----- 16.7/16.7 MB 220.6 MB/s 0:00:00
  Downloading protobuf-7.35.1-cp310-abi3-manylinux2014_x86_64.whl (327 kB)
  Downloading flatbuffers-25.12.19-py2.py3-none-any.whl (26 kB)
  Downloading packaging-26.2-py3-none-any.whl (100 kB)
Installing collected packages: flatbuffers, protobuf, pathvalidate, packaging, num
py, onnxruntime, piper-tts
Successfully installed flatbuffers-25.12.19 numpy-2.5.1 onnxruntime-1.27.0 packagi
ng-26.2 pathvalidate-3.3.1 piper-tts-1.4.2 protobuf-7.35.1
ubuntu@ip-172-31-18-142:~$

```

ls -lh /opt/piper/bin/python

```

ubuntu@ip-172-31-18-142:~$ ls -lh /opt/piper/bin/python
lrwxrwxrwx 1 root root 7 Jul 14 13:30 /opt/piper/bin/python -> python3
ubuntu@ip-172-31-18-142:~$

```

/opt/piper/bin/python -m piper --help

```

ubuntu@ip-172-31-18-142:~$ /opt/piper/bin/python -m piper --help
2026-07-14 13:31:19.464026497 [W:onnxruntime:Default, device_discovery.cc:283 GetG
puDevices] Failed to detect devices under "/sys/class/drm/card0": device_discovery
.cc:93 ReadFileContents Failed to open file: "/sys/class/drm/card0/device/vendor"
usage: __main__.py [-h] -m MODEL [-c CONFIG] [-i INPUT_FILE] [-f OUTPUT_FILE]
                  [-d OUTPUT_DIR] [--output-dir-naming {timestamp,text}]
                  [--output-raw] [-s SPEAKER] [--length-scale LENGTH_SCALE]
                  [--noise-scale NOISE_SCALE] [--noise-w-scale NOISE_W_SCALE]
                  [--cuda] [--sentence-silence SENTENCE_SILENCE]
                  [--volume VOLUME] [--no-normalize] [--data-dir DATA_DIR]
                  [--debug]

options:

```

2. Descargar la voz española

Piper necesita dos archivos por voz: el modelo .onnx y su configuración .onnx.json.

```
sudo mkdir -p /opt/piper/voices
```

```
sudo /opt/piper/bin/python \
```

```
-m piper.download_voices \
```

```
es_ES-davefx-medium \
```

```
--data-dir /opt/piper/voices
```

```
ubuntu@ip-172-31-18-142:~$ sudo mkdir -p /opt/piper/voices

sudo /opt/piper/bin/python \
-m piper.download_voices \
  es_ES-davefx-medium \
  --data-dir /opt/piper/voices
2026-07-14 13:32:31.984164153 [W:onnxruntime:Default, device_discovery.cc:283 GetG
puDevices] Failed to detect devices under "/sys/class/drm/card0": device_discovery
.cc:93 ReadFileContents Failed to open file: "/sys/class/drm/card0/device/vendor"
INFO: __main__:Downloaded: es_ES-davefx-medium
ubuntu@ip-172-31-18-142:~$
```

Asigna permisos a Asterisk:

```
sudo chown -R asterisk:asterisk /opt/piper/voices
```

```
sudo chmod -R 755 /opt/piper
```

```
INFO: __main__:Downloaded: es_ES-davefx-medium
ubuntu@ip-172-31-18-142:~$ sudo chown -R asterisk:asterisk /opt/piper/voices
sudo chmod -R 755 /opt/piper
ubuntu@ip-172-31-18-142:~$ ls -lh /opt/piper/voices
total 61M
-rwxr-xr-x 1 asterisk asterisk 61M Jul 14 13:32 es_ES-davefx-medium.onnx
-rwxr-xr-x 1 asterisk asterisk 4.8K Jul 14 13:32 es_ES-davefx-medium.onnx.json
ubuntu@ip-172-31-18-142:~$
```

4. Convertir el audio al formato de Asterisk

```
sudo -u asterisk sox \
```

```
/tmp/prueba-piper.wav \
```

```
-r 8000 \
```

-c 1 \

-b 16 \

-e signed-integer \

/var/lib/asterisk/sounds/custom/citas/prueba-piper.wav

```
ubuntu@ip-172-31-18-142:/tmp$ sudo -u asterisk sox \  
  /tmp/prueba-piper.wav \  
  -r 8000 \  
  -c 1 \  
  -b 16 \  
  -e signed-integer \  
  /var/lib/asterisk/sounds/custom/citas/prueba-piper.wav  
sox WARN rate: rate clipped 1 samples; decrease volume?  
sox WARN dither: dither clipped 1 samples; decrease volume?  
ubuntu@ip-172-31-18-142:/tmp$
```

soxi /var/lib/asterisk/sounds/custom/citas/prueba-piper.wav

```

Input File      : '/var/lib/asterisk/sounds/custom/citas/prueba-piper.wav'
Channels       : 1
Sample Rate    : 8000
Precision      : 16-bit
Duration       : 00:00:02.59 = 20712 samples ~ 194.175 CDDA sectors
File Size      : 41.5k
Bit Rate       : 128k
Sample Encoding: 16-bit Signed Integer PCM

ubuntu@ip-172-31-18-142:/tmp$ sudo -u asterisk sox \
    /tmp/prueba-piper.wav \
    -r 8000 \
    -c 1 \
    -b 16 \
    -e signed-integer \
    /var/lib/asterisk/sounds/custom/citas/prueba-piper.wav
sox WARN rate: rate clipped 1 samples; decrease volume?
sox WARN dither: dither clipped 1 samples; decrease volume?
ubuntu@ip-172-31-18-142:/tmp$ 4. Convertir el audio al formato de Asterisk
4.: command not found
ubuntu@ip-172-31-18-142:/tmp$ soxi /var/lib/asterisk/sounds/custom/citas/prueba-piper.wav

Input File      : '/var/lib/asterisk/sounds/custom/citas/prueba-piper.wav'
Channels       : 1
Sample Rate    : 8000
Precision      : 16-bit
Duration       : 00:00:02.59 = 20712 samples ~ 194.175 CDDA sectors
File Size      : 41.5k
Bit Rate       : 128k
Sample Encoding: 16-bit Signed Integer PCM

ubuntu@ip-172-31-18-142:/tmp$ █

```

5. Verificación final

```
echo "===== PYTHON PIPER ====="
```

```
/opt/piper/bin/python --version
```

```
echo "===== PIPER ====="
```

```
/opt/piper/bin/python -m piper --help | head
```

```
echo "===== VOZ ====="
```

```
ls -lh /opt/piper/voices/
```



```
echo "===== AUDIO ASTERISK ====="
```

```
ls -lh /var/lib/asterisk/sounds/custom/citas/prueba-piper.wav
```

```
ubuntu@ip-172-31-18-142:/tmp$ echo "===== PYTHON PIPER ====="
/opt/piper/bin/python --version

echo "===== PIPER ====="
/opt/piper/bin/python -m piper --help | head

echo "===== VOZ ====="
ls -lh /opt/piper/voices/

echo "===== AUDIO ASTERISK ====="
ls -lh /var/lib/asterisk/sounds/custom/citas/prueba-piper.wav
===== PYTHON PIPER =====
Python 3.12.3
===== PIPER =====
2026-07-14 13:37:46.486541656 [W:onnxruntime:Default, device_discovery.cc:283 GetG
puDevices] Failed to detect devices under "/sys/class/drm/card0": device_discovery
.cc:93 ReadFileContents Failed to open file: "/sys/class/drm/card0/device/vendor"
usage: __main__.py [-h] -m MODEL [-c CONFIG] [-i INPUT_FILE] [-f OUTPUT_FILE]
                  [-d OUTPUT_DIR] [--output-dir-naming {timestamp,text}]
                  [--output-raw] [-s SPEAKER] [--length-scale LENGTH_SCALE]
                  [--noise-scale NOISE_SCALE] [--noise-w-scale NOISE_W_SCALE]
                  [--cuda] [--sentence-silence SENTENCE_SILENCE]
                  [--volume VOLUME] [--no-normalize] [--data-dir DATA_DIR]
                  [--debug]

options:
  -h, --help                show this help message and exit
===== VOZ =====
total 61M
-rwxr-xr-x 1 asterisk asterisk 61M Jul 14 13:32 es_ES-davefx-medium.onnx
-rwxr-xr-x 1 asterisk asterisk 4.8K Jul 14 13:32 es_ES-davefx-medium.onnx.json
===== AUDIO ASTERISK =====
-rw-rw-r-- 1 asterisk asterisk 41K Jul 14 13:36 /var/lib/asterisk/sounds/custom/ci
tas/prueba-piper.wav
ubuntu@ip-172-31-18-142:/tmp$
```

1. Crear la cuenta SIP 1001
2. `sudo cp /etc/asterisk/pjsip.conf \`
3. `/etc/asterisk/pjsip.conf.backup-voicebot`
- 4.
5. `sudo cp /etc/asterisk/extensions.conf \`

6. /etc/asterisk/extensions.conf.backup-voicebot

sudo nano /etc/asterisk/pjsip_voicebot.conf

; =====

; CUENTA SIP 1001 PARA MICROSIP

; =====

[1001-auth]

type=auth

auth_type=userpass

username=1001

password=VoiceBot1001-2026!

[1001-aor]

type=aor

max_contacts=1

remove_existing=yes

qualify_frequency=30

[1001]

type=endpoint

context=from-voicebot-users

disallow=all

allow=ulaw

allow=alaw

auth=1001-auth

aors=1001-aor

callerid=MicroSIP 1001 <1001>

dtmf_mode=rfc4733

direct_media=no

rtp_symmetric=yes

force_rport=yes

rewrite_contact=yes

2. Incluir el archivo en PJSIP

**grep -qF '#include pjsip_voicebot.conf' **

**/etc/asterisk/pjsip.conf **

**|| echo '#include pjsip_voicebot.conf' **

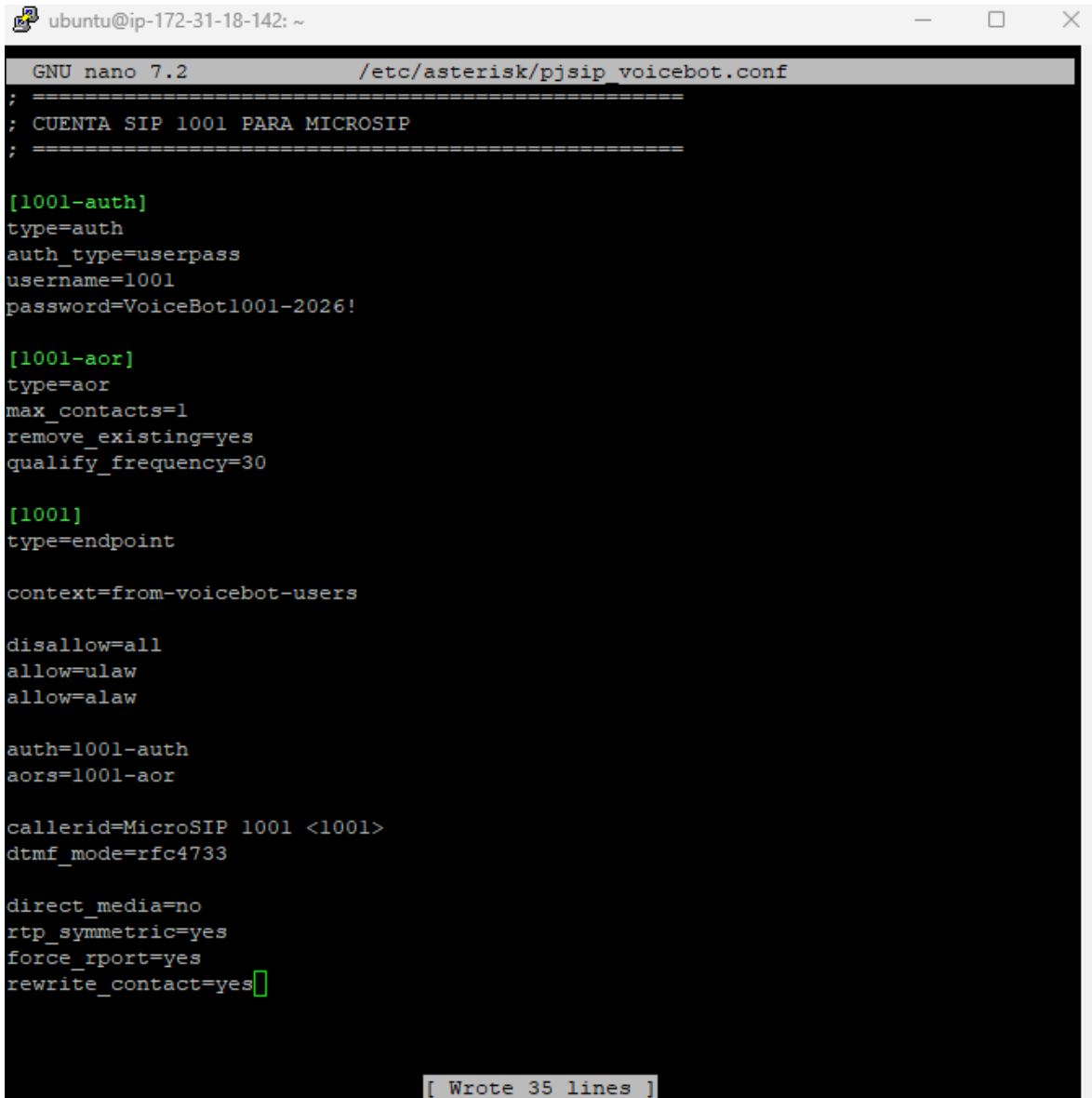
| sudo tee -a /etc/asterisk/pjsip.conf

```
Ctrl: command not found
ubuntu@ip-172-31-18-142:~$ grep -qF '#include pjsip_voicebot.conf' \
/etc/asterisk/pjsip.conf \
|| echo '#include pjsip_voicebot.conf' \
| sudo tee -a /etc/asterisk/pjsip.conf
#include pjsip_voicebot.conf
ubuntu@ip-172-31-18-142:~$
```

tail -5 /etc/asterisk/pjsip.conf

```
ubuntu@ip-172-31-18-142:~$ tail -5 /etc/asterisk/pjsip.conf
use_callerid_contact=no
keep_alive_interval=90
#include pjsip_custom_post.conf

#include pjsip_voicebot.conf
ubuntu@ip-172-31-18-142:~$
```



```
ubuntu@ip-172-31-18-142: ~
GNU nano 7.2 /etc/asterisk/pjsip_voicebot.conf
; =====
; CUENTA SIP 1001 PARA MICROSIP
; =====

[1001-auth]
type=auth
auth_type=userpass
username=1001
password=VoiceBot1001-2026!

[1001-aor]
type=aor
max_contacts=1
remove_existing=yes
qualify_frequency=30

[1001]
type=endpoint

context=from-voicebot-users

disallow=all
allow=ulaw
allow=alaw

auth=1001-auth
aors=1001-aor

callerid=MicroSIP 1001 <1001>
dtmf_mode=rfc4733

direct_media=no
rtp_symmetric=yes
force_rport=yes
rewrite_contact=yes
[ Wrote 35 lines ]
```

Crear los números 700 y 701

```
sudo nano /etc/asterisk/extensions_voicebot.conf
```

```
ubuntu@ip-172-31-18-142:~$ sudo nano /etc/asterisk/extensions_voicebot.conf
```

```
;=====
; EXTENSIONES DEL PROYECTO VOICEBOT
;=====
```

. Crear un contexto separado para Voicebot

Usaremos un contexto independiente llamado:

voicebot-custom

y lo conectaremos con from-internal-custom.

Ejecuta este bloque completo:

. Crear un contexto separado para Voicebot

Usaremos un contexto independiente llamado:

voicebot-custom

y lo conectaremos con from-internal-custom.

Ejecuta este bloque completo:

Aplicar permisos

```
ubuntu@ip-172-31-18-142:~$ sudo chown root:asterisk \
/etc/asterisk/extensions_custom.conf

sudo chmod 640 \
/etc/asterisk/extensions_custom.conf
ubuntu@ip-172-31-18-142:~$
```

Verificación:

```
sudo ls -lh \
```

```
/etc/asterisk/extensions_custom.conf
```

```
ubuntu@ip-172-31-18-142:~$ sudo ls -lh \
/etc/asterisk/extensions_custom.conf
-rw-r----- 1 root asterisk 495 Jul 14 21:05 /etc/asterisk/extensions_custom.conf
ubuntu@ip-172-31-18-142:~$
```

Revisar el contenido creado

```
ubuntu@ip-172-31-18-142:~$ sudo grep -nA12 -B3 \
-E 'from-internal-custom|voicebot-custom|exten => \*70' \
/etc/asterisk/extensions_custom.conf
1-
2:[from-internal-custom]
3:include => voicebot-custom
4-
5-
6-; =====
7-; PROYECTO VOICEBOT
8-; =====
9-
10:[voicebot-custom]
11-
12-; -----
13-; *70 - PRUEBA DE VOZ PIPER
14-; -----
15-
16:exten => *70,1,NoOp(Prueba de voz generada con Piper)
17- same => n,Answer()
18- same => n,Wait(1)
19- same => n,Playback(custom/citas/prueba-piper)
20- same => n,Hangup()
21-
```

Resultado de la verificación

Asterisk: activo

Archivo de audio: existe

Canales: 1

Frecuencia: 8000 Hz

Profundidad: 16 bits

Dialplan *70: cargado

Asterisk reconoce las cinco instrucciones:

1. NoOp(Prueba de voz generada con Piper)
2. Answer()
3. Wait(1)
4. Playback(custom/citas/prueba-piper)

5. Hangup()

Verificación de voz

```

3458991acb3a900bb9004", algorithm=MD5, cnonce="a2a33847299345379b8f442f4020436a
paque="4cf0c4a809c0f070", qop=auth, nc=00000001
o: <sip:*70@50.19.179.176>
2026-07-14 21:13:50] VERBOSE[7577][C-00000002] pbx.c: Executing [*70@from-inter
1] NoOp("PJSIP/1001-00000001", "Prueba de voz generada con Piper") in new stack
2026-07-14 21:13:50] VERBOSE[7577][C-00000002] pbx.c: Executing [*70@from-inter
2] Answer("PJSIP/1001-00000001", "") in new stack
o: <sip:*70@50.19.179.176>;tag=0249c66f-d944-4592-9460-ee7999e7bbb6
o: <sip:*70@50.19.179.176>;tag=0249c66f-d944-4592-9460-ee7999e7bbb6
2026-07-14 21:13:50] VERBOSE[7577][C-00000002] pbx.c: Executing [*70@from-inter
3] Wait("PJSIP/1001-00000001", "1") in new stack
o: <sip:*70@50.19.179.176>;tag=0249c66f-d944-4592-9460-ee7999e7bbb6
2026-07-14 21:13:51] VERBOSE[7577][C-00000002] pbx.c: Executing [*70@from-inter
4] Playback("PJSIP/1001-00000001", "custom/citas/prueba-piper") in new stack
2026-07-14 21:13:51] VERBOSE[7577][C-00000002] file.c: <PJSIP/1001-00000001> Pl
ay 'custom/citas/prueba-piper.slin' (language 'en')
o: <sip:*70@50.19.179.176>;tag=0249c66f-d944-4592-9460-ee7999e7bbb6
2026-07-14 21:13:54] VERBOSE[7577][C-00000002] pbx.c: Executing [*70@from-inter
5] Hangup("PJSIP/1001-00000001", "") in new stack
2026-07-14 21:13:54] VERBOSE[7577][C-00000002] pbx.c: Spawn extension (from-int
al, *70, 5) exited non-zero on 'PJSIP/1001-00000001'
o: <sip:*70@50.19.179.176>;tag=0249c66f-d944-4592-9460-ee7999e7bbb6
o: <sip:*70@50.19.179.176>;tag=0249c66f-d944-4592-9460-ee7999e7bbb6
o: <sip:*70@50.19.179.176>;tag=0249c66f-d944-4592-9460-ee7999e7bbb6
o: <sip:*70@50.19.179.176>;tag=0249c66f-d944-4592-9460-ee7999e7bbb6
o: <sip:*70@50.19.179.176>;tag=0249c66f-d944-4592-9460-ee7999e7bbb6
o: <sip:*70@50.19.179.176>;tag=0249c66f-d944-4592-9460-ee7999e7bbb6
o: <sip:*70@50.19.179.176>;tag=0249c66f-d944-4592-9460-ee7999e7bbb6
com: <sip:*70@50.19.179.176>;tag=0249c66f-d944-4592-9460-ee7999e7bbb6
com: <sip:*70@50.19.179.176>;tag=0249c66f-d944-4592-9460-ee7999e7bbb6
ubuntu@ip-172-31-18-142:~$

```

Como se puede apreciar al llamar al anexo *70 el agente me comunica soy el agente de reserva

Etapas 2 — Sistema de citas con AGI y DTMF

1. Instalar SQLite y comprobar dependencias

```

ubuntu@ip-172-31-18-142:~$ sudo apt update

sudo apt install -y \
    sqlite3 \
    python3 \
    sox

```

- 2.

Verificación

```
echo "===== PYTHON ====="
```

```
python3 --version
```

```
echo "===== SQLITE ====="
```

```
sqlite3 --version
```

```
echo "===== SOX ====="
```

```
sox --version
```

```
echo "===== PIPER ====="
```

```
/opt/piper/bin/python -m piper --help | head -10
```

Crear la estructura del sistema de citas

```
sudo install -d \
```

```
-o asterisk \
```

```
-g asterisk \
```

```
/var/lib/asterisk/citas
```

```
sudo install -d \
```

```
-o asterisk \
```

```
-g asterisk \
```

```
/var/lib/asterisk/sounds/custom/citas
```

```
sudo touch /var/log/asterisk/citas-agi.log
```



```
sudo chown asterisk:asterisk \
```

```
/var/log/asterisk/citas-agi.log
```

```
sudo chmod 640 \
```

```
/var/log/asterisk/citas-agi.log
```

```
ubuntu@ip-172-31-18-142:~$ sudo install -d \
-o asterisk \
-g asterisk \
/var/lib/asterisk/citas

sudo install -d \
-o asterisk \
-g asterisk \
/var/lib/asterisk/sounds/custom/citas

sudo touch /var/log/asterisk/citas-agi.log

sudo chown asterisk:asterisk \
/var/log/asterisk/citas-agi.log

sudo chmod 640 \
/var/log/asterisk/citas-agi.log
ubuntu@ip-172-31-18-142:~$
```

Verificación

```
ubuntu@ip-172-31-18-142:~$ echo "===== DIRECTORIO CITAS ====="
sudo ls -ld /var/lib/asterisk/citas

echo "===== DIRECTORIO AUDIOS ====="
sudo ls -ld /var/lib/asterisk/sounds/custom/citas

echo "===== LOG ====="
sudo ls -lh /var/log/asterisk/citas-agi.log
===== DIRECTORIO CITAS =====
drwxr-xr-x 2 asterisk asterisk 4096 Jul 14 21:22 /var/lib/asterisk/citas
===== DIRECTORIO AUDIOS =====
drwxr-xr-x 2 asterisk asterisk 4096 Jul 14 13:36 /var/lib/asterisk/sounds/custom/
itas
===== LOG =====
-rw-r----- 1 asterisk asterisk 0 Jul 14 21:23 /var/log/asterisk/citas-agi.log
ubuntu@ip-172-31-18-142:~$
```

Crear la base de datos SQLite

```
ubuntu@ip-172-31-18-142:~$ sudo -u asterisk sqlite3 \
/var/lib/asterisk/citas/citas.db <<'SQL'
```

```
PRAGMA foreign_keys = ON;
```

```
CREATE TABLE IF NOT EXISTS citas (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    telefono TEXT NOT NULL,
    nombre TEXT NOT NULL,
    fecha_texto TEXT NOT NULL,
    hora_texto TEXT NOT NULL,
    motivo TEXT NOT NULL,
    estado TEXT NOT NULL DEFAULT 'pendiente',
    actualizado_en TEXT
);
```

```
CREATE TABLE IF NOT EXISTS resultados (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    cita_id INTEGER NOT NULL,
    telefono TEXT,
    digito TEXT,
    accion TEXT NOT NULL,
    fecha_registro TEXT NOT NULL,
    FOREIGN KEY(cita_id) REFERENCES citas(id)
);
```

```
INSERT INTO citas (
```

```
    telefono,
    nombre,
    fecha_texto,
    hora_texto,
    motivo,
    estado
)
SELECT
    '1001',
    'Rodrigo',
    'veinte de julio',
    'diez treinta de la mañana',
    'una cita de prueba',
    'pendiente'
WHERE NOT EXISTS (
SQL WHERE telefono = '1001'
ubuntu@ip-172-31-18-142:~$
```

```
sudo -u asterisk sqlite3 \
```

```
/var/lib/asterisk/citas/citas.db <<'SQL'
```

```
PRAGMA foreign_keys = ON;
```

```
CREATE TABLE IF NOT EXISTS citas (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    telefono TEXT NOT NULL,  
    nombre TEXT NOT NULL,  
    fecha_texto TEXT NOT NULL,  
    hora_texto TEXT NOT NULL,  
    motivo TEXT NOT NULL,  
    estado TEXT NOT NULL DEFAULT 'pendiente',  
    actualizado_en TEXT  
);
```

```
CREATE TABLE IF NOT EXISTS resultados (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    cita_id INTEGER NOT NULL,  
    telefono TEXT,  
    digito TEXT,  
    accion TEXT NOT NULL,  
    fecha_registro TEXT NOT NULL,  
    FOREIGN KEY(cita_id) REFERENCES citas(id)  
);
```

```
INSERT INTO citas (  
    telefono,
```

```
nombre,  
fecha_texto,  
hora_texto,  
motivo,  
estado  
)  
SELECT  
    '1001',  
    'Rodrigo',  
    'veinte de julio',  
    'diez treinta de la mañana',  
    'una cita de prueba',  
    'pendiente'  
WHERE NOT EXISTS (  
    SELECT 1  
    FROM citas  
    WHERE telefono = '1001'  
);
```

SQL

Verificar tablas

```
sudo -u asterisk sqlite3 \  
/var/lib/asterisk/citas/citas.db \  
".tables"
```

```
buntu@ip-172-31-18-142:~$ sudo -u asterisk sqlite3 \
/var/lib/asterisk/citas/citas.db \
.headers on" \
.mode column" \
SELECT * FROM citas;"
d telefono nombre fecha_texto hora_texto motivo
estado actualizado_en
-----
1001 Rodrigo veinte de julio diez treinta de la mañana una cita de pru
ba pendiente
buntu@ip-172-31-18-142:~$ sudo ls -lh \
/var/lib/asterisk/citas/citas.db
-rw-r--r-- 1 asterisk asterisk 16K Jul 14 21:24 /var/lib/asterisk/citas/citas.db
buntu@ip-172-31-18-142:~$
```

```
echo "==== SQLITE ====="
sqlite3 --version

echo "==== SOX ====="
sox --version

echo "==== PIPER ====="
/opt/piper/bin/python -m piper --help | head -10
==== PYTHON =====
Python 3.12.3
==== SQLITE =====
3.45.1 2024-01-30 16:01:20 e876e51a0ed5c5b3126f52e532044363a014bc594cfefa87ffb5b82
257ccaltl (64-bit)
==== SOX =====
sox: SoX v14.4.2
==== PIPER =====
2026-07-14 21:21:32.768109553 [W:onnxruntime:Default, device_discovery.cc:283 GetG
puDevices] Failed to detect devices under "/sys/class/drm/card0": device_discovery
.cc:93 ReadFileContents Failed to open file: "/sys/class/drm/card0/device/vendor"
usage: __main__.py [-h] -m MODEL [-c CONFIG] [-i INPUT_FILE] [-f OUTPUT_FILE]
                  [-d OUTPUT_DIR] [--output-dir-naming {timestamp,text}]
                  [--output-raw] [-s SPEAKER] [--length-scale LENGTH_SCALE]
                  [--noise-scale NOISE_SCALE] [--noise-w-scale NOISE_W_SCALE]
                  [--cuda] [--sentence-silence SENTENCE_SILENCE]
                  [--volume VOLUME] [--no-normalize] [--data-dir DATA_DIR]
                  [--debug]

options:
  -h, --help            show this help message and exit
```

. Crear el archivo de configuración del AGI

```
sudo tee /etc/asterisk/citas.ini >/dev/null <<'EOF'
```

[general]

database=/var/lib/asterisk/citas/citas.db

piper_python=/opt/piper/bin/python

piper_model=/opt/piper/voices/es_ES-davefx-medium.onnx

sounds_dir=/var/lib/asterisk/sounds/custom/citas

log_file=/var/log/asterisk/citas-agi.log

EOF

```
ubuntu@ip-172-31-18-142:~$ sudo tee /etc/asterisk/citas.ini >/dev/null <<'EOF'
[general]

database=/var/lib/asterisk/citas/citas.db

piper_python=/opt/piper/bin/python
piper_model=/opt/piper/voices/es_ES-davefx-medium.onnx

sounds_dir=/var/lib/asterisk/sounds/custom/citas

log_file=/var/log/asterisk/citas-agi.log
EOF
ubuntu@ip-172-31-18-142:~$ sudo chown root:asterisk \
/etc/asterisk/citas.ini

sudo chmod 640 \
/etc/asterisk/citas.ini
ubuntu@ip-172-31-18-142:~$
```

5. Crear el script AGI

```
ubuntu@ip-172-31-18-142:/var/lib/asterisk/agi-bin$ sudo chown asterisk:asterisk \
/var/lib/asterisk/agi-bin/citas.py

sudo chmod 750 \
/var/lib/asterisk/agi-bin/citas.py
ubuntu@ip-172-31-18-142:/var/lib/asterisk/agi-bin$
```

```

ubuntu@ip-172-31-18-142:/var/lib/asterisk/agi-bin$ sudo tee /var/lib/asterisk/agi-bin/citas.py >/dev/null <<'PYTHON'
#!/usr/bin/env python3

from __future__ import annotations

import configparser
import logging
import re
import sqlite3
import subprocess
import sys
import tempfile
from datetime import datetime
from pathlib import Path

CONFIG_PATH = "/etc/asterisk/citas.ini"

def read_agi_environment() -> dict[str, str]:
    """
    Lee las variables enviadas por Asterisk al iniciar AGI.
    """
    environment: dict[str, str] = {}
ubuntu@ip-172-31-18-142:/var/lib/asterisk/agi-bin$ sudo ls -lh \
/var/lib/asterisk/agi-bin/citas.py
-rwxr-x--- 1 asterisk asterisk 7.9K Jul 14 21:27 /var/lib/asterisk/agi-bin/citas.py
ubuntu@ip-172-31-18-142:/var/lib/asterisk/agi-bin$

```

7. Crear el código telefónico *71

```
ubuntu@ip-172-31-18-142:/var/lib/asterisk/agi-bin$ CUSTOM_FILE="/etc/asterisk/extensions_
om.conf"

if ! sudo grep -q \
'^[[:space:]]*include[[:space:]]*=>[[:space:]]*voicebot-agi' \
"$CUSTOM_FILE"; then

    sudo sed -i \
    '/^[[:space:]]*include => voicebot-agi' \
    "$CUSTOM_FILE"
fi

if ! sudo grep -q \
'^\[voicebot-agi\]$' \
"$CUSTOM_FILE"; then

    sudo tee -a "$CUSTOM_FILE" >/dev/null <<'EOF'

; =====
; SISTEMA AGI DE CITAS
; =====

[voicebot-agi]

; -----
; *71 - MENÚ DE CITAS CON DTMF
; -----

exten => *71,1,NoOp(Sistema AGI de citas)
same => n,Answer()
same => n,Wait(1)
same => n,AGI(citas.py)
same => n,Verbose(1,AGI citas finalizó con estado ${AGISTATUS})
same => n,Hangup()
```

CUSTOM_FILE="/etc/asterisk/extensions_custom.conf"

```
if ! sudo grep -q \
'^[[:space:]]*include[[:space:]]*=>[[:space:]]*voicebot-agi' \
"$CUSTOM_FILE"; then

    sudo sed -i \
    '/^[[:space:]]*include => voicebot-agi' \
    "$CUSTOM_FILE"
fi

if ! sudo grep -q \
'^\[voicebot-agi\]$' \
"$CUSTOM_FILE"; then

    sudo tee -a "$CUSTOM_FILE" >/dev/null <<'EOF'
```



```
; =====  
; SISTEMA AGI DE CITAS  
; =====
```

[voicebot-agi]

```
; -----  
; *71 - MENÚ DE CITAS CON DTMF  
; -----
```

```
exten => *71,1,NoOp(Sistema AGI de citas)  
same => n,Answer()  
same => n,Wait(1)  
same => n,AGI(citas.py)  
same => n,Verbose(1,AGI citas finalizó con estado ${AGISTATUS})  
same => n,Hangup()
```

EOF

Fi

Aplicar permisos

```
sudo chown root:asterisk \  
/etc/asterisk/extensions_custom.conf
```

```
sudo chmod 640 \  
/etc/asterisk/extensions_custom.conf
```

```
ubuntu@ip-172-31-18-142:/var/lib/asterisk/agi-bin$ sudo grep -nA12 -B3 \
-E 'voicebot-agi|exten => \*71' \
/etc/asterisk/extensions_custom.conf
1-
2-[from-internal-custom]
3:include => voicebot-agi
4:include => voicebot-custom
5-
6-
7-; =====
8-; PROYECTO VOICEBOT
9-; =====
10-
11-[voicebot-custom]
12-
13-; -----
14-; *70 - PRUEBA DE VOZ PIPER
15-; -----
16-
17-
18-
19-
20-
21-
22-
23-
24-; SISTEMA AGI DE CITAS
25-; =====
26-;
27-
28-[voicebot-agi]
29-
30-; -----
31-; *71 - MENÚ DE CITAS CON DTMF
32-; -----
33-
34:exten => *71,1,NoOp(Sistema AGI de citas)
35- same => n,Answer()
36- same => n,Wait(1)
37- same => n,AGI(citas.py)
38- same => n,Verbose(1,AGI citas finalizó con estado ${AGISTATUS})
39- same => n,Hangup()
40-
ubuntu@ip-172-31-18-142:/var/lib/asterisk/agi-bin$
```

10. Verificar que *71 esté cargado

```
ubuntu@ip-172-31-18-142:/var/lib/asterisk/agi-bin$ sudo asterisk -rx \
'dialplan show *71@voicebot-agi'
[ Context 'voicebot-agi' created by 'pbx_config' ]
  '*71' =>
          1. NoOp(Sistema AGI de citas) [extensions_custom.conf:34]
          2. Answer() [extensions_custom.conf:35]
          3. Wait(1) [extensions_custom.conf:36]
          4. AGI(citas.py) [extensions_custom.conf:37]
          5. Verbose(1,AGI citas finalizó con estado ${AGISTATUS}) [extensions_cus
om.conf:38]
          6. Hangup() [extensions_custom.conf:39]

-- 1 extension (6 priorities) in 1 context. --
ubuntu@ip-172-31-18-142:/var/lib/asterisk/agi-bin$
```

11. Verificación general antes de llamarecho "===== ASTERISK ====="

sudo systemctl is-active asterisk

```
echo "===== CONTACTO 1001 ====="

sudo asterisk -rx "pjsip show contacts" \

| grep 1001 || true
```

```
echo "===== BASE DE DATOS ====="

sudo -u asterisk sqlite3 \

/var/lib/asterisk/citas/citas.db \

.headers on \

.mode column \

"SELECT id,telefono,nombre,estado FROM citas;"
```

```
echo "===== SCRIPT AGI ====="

sudo ls -lh \

/var/lib/asterisk/agi-bin/citas.py
```

```
echo "===== CONFIGURACIÓN ====="

sudo ls -lh \

/etc/asterisk/citas.ini
```

```
echo "===== DIALPLAN *71 ====="

sudo asterisk -rx \

"dialplan show *71@voicebot-agi"
```

```
buntu@ip-172-31-18-142:/var/lib/asterisk/agi-bin$ sudo asterisk -rx \
dialplan show *71@voicebot-agi"
Context 'voicebot-agi' created by 'pbx_config' ]
'*71' =>
1. NoOp(Sistema AGI de citas) [extensions_custom.conf:38]
2. Answer() [extensions_custom.conf:38]
3. Wait(1) [extensions_custom.conf:38]
4. AGI(citas.py) [extensions_custom.conf:38]
5. Verbose(1,AGI citas finalizó con estado ${AGISTATUS}) [extensions_custom.conf:38]
6. Hangup() [extensions_custom.conf:38]

= 1 extension (6 priorities) in 1 context. ==
buntu@ip-172-31-18-142:/var/lib/asterisk/agi-bin$ echo "===== ASTERISK ====="
udo systemctl is-active asterisk

cho "===== CONTACTO 1001 ====="
udo asterisk -rx "pjsip show contacts" \
grep 1001 || true
```

```
cho "===== BASE DE DATOS ====="
udo -u asterisk sqlite3 \
/var/lib/asterisk/citas/citas.db \
'.headers on' \
'.mode column' \
'SELECT id,telefono,nombre,estado FROM citas;"

cho "===== SCRIPT AGI ====="
udo ls -lh \
/var/lib/asterisk/agi-bin/citas.py

cho "===== CONFIGURACIÓN ====="
udo ls -lh \
/etc/asterisk/citas.ini

cho "===== DIALPLAN *71 ====="
udo asterisk -rx \
'dialplan show *71@voicebot-agi"
===== ASTERISK =====
active
===== CONTACTO 1001 =====
Contact: 1001/sip:1001@179.60.64.152:61170;ob 2ec0ed8b07 Avail 136.069
===== BASE DE DATOS =====
id telefono nombre estado
-----
1 1001 Rodrigo pendiente
===== SCRIPT AGI =====
-rwxr-x--- 1 asterisk asterisk 7.9K Jul 14 21:27 /var/lib/asterisk/agi-bin/citas.py
===== CONFIGURACIÓN =====
-rw-r----- 1 root asterisk 236 Jul 14 21:26 /etc/asterisk/citas.ini
===== DIALPLAN *71 =====
[ Context 'voicebot-agi' created by 'pbx_config' ]
'*71' =>
1. NoOp(Sistema AGI de citas) [extensions_custom.conf:34]
2. Answer() [extensions_custom.conf:35]
3. Wait(1) [extensions_custom.conf:36]
4. AGI(citas.py) [extensions_custom.conf:37]
5. Verbose(1,AGI citas finalizó con estado ${AGISTATUS}) [extensions_custom.conf:38]
6. Hangup() [extensions_custom.conf:39]

= 1 extension (6 priorities) in 1 context. ==
buntu@ip-172-31-18-142:/var/lib/asterisk/agi-bin$
```

Sesión de prueba con la reserva

```
ubuntu@ip-172-31-18-142:~$ sudo -u asterisk sqlite3 \
/var/lib/asterisk/citas/citas.db <<'SQL'

UPDATE citas
SET
    nombre = 'Francisco',
    fecha_texto = 'diecisiete de julio',
    hora_texto = 'diez de la mañana',
    motivo = 'con el doctor Sebastián Pizarro en la clínica universitaria',
    estado = 'pendiente',
    actualizado_en = NULL
WHERE telefono = '1001';

INSERT INTO citas (
    telefono,
    nombre,
    fecha_texto,
    hora_texto,
    motivo,
    estado,
    actualizado_en
)
SELECT
    '1001',
    'Francisco',
    'diecisiete de julio',
    'diez de la mañana',
    'con el doctor Sebastián Pizarro en la clínica universitaria',
    'pendiente',
    NULL
WHERE NOT EXISTS (
    SELECT 1
    FROM citas
    WHERE telefono = '1001'
);

SQL
ubuntu@ip-172-31-18-142:~$ sudo -u asterisk sqlite3 \
/var/lib/asterisk/citas/citas.db \
.headers on \
.mode column \
"SELECT
id,
```

Se verifica que la reserva esta pendiente

```
ubuntu@ip-172-31-18-142:~$ sudo -u asterisk sqlite3 \
/var/lib/asterisk/citas/citas.db \
.headers on \
.mode column \
"SELECT
id,
telefono,
nombre,
fecha_texto,
hora_texto,
motivo,
estado
FROM citas
WHERE telefono = '1001';"
id telefono nombre fecha_texto hora_texto motivo estado
--
1 1001 Francisco diecisiete de julio diez de la mañana con el doctor Sebastián Pizarro en la clínica universitaria pendiente
ubuntu@ip-172-31-18-142:~$
```

Conclusión

La implementación de la central telefónica permitió desplegar una solución basada en Asterisk y FreePBX en un servidor cloud, configurando extensiones SIP y softphones para realizar las pruebas de comunicación.

El script AGI desarrollado permitió consultar los datos de las citas, reproducir mensajes personalizados con el nombre del paciente, especialidad, fecha y hora, y capturar respuestas mediante teclas DTMF. La opción 1 permitió confirmar la cita, la opción 2 cancelarla y la opción 3 registrar una solicitud de reprogramación.

Además, el sistema actualizó automáticamente el estado de las citas y almacenó los resultados en archivos CSV y registros de auditoría. También se implementó la generación de llamadas automáticas y el control de intentos para pacientes que no respondieron.

En conclusión, la solución cumplió con los principales requerimientos del laboratorio, permitiendo automatizar el proceso de confirmación de citas, disminuir el trabajo manual y mantener un registro verificable de cada llamada y respuesta